

A Synopsis Report

on

A TinyML Powered Tactile Environmental Alerting System for Deaf & Hard of Hearing

FINAL YEAR ENGINEERING

Final Project

of

E and TC Department

Submitted by

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Chapter 1

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Chapter 2

Problem Statement

- Problem Statement "Individuals who are Deaf or Hard of Hearing (DHH) are functionally vulnerable to critical, low amplitude or sudden acoustic warnings in their everyday environments, such as fire alarms (2-4kHz), vehicle horns, or emergency sirens—until potential physical hazards or accidents occur.

Existing wall-mounted visual alerting systems and traditional hearing aids suffer from:

- high latency in complex environments
- line-of-sight visual limitations
- network dependence and unsuitability for profound deafness.

This project solves the problem by deploying an ultra-low-power, wearable ESP32 TinyML node that detects acoustic anomalies and critical environmental sounds on-device in <40 ms and alerts the user immediately via localized haptic feedback."

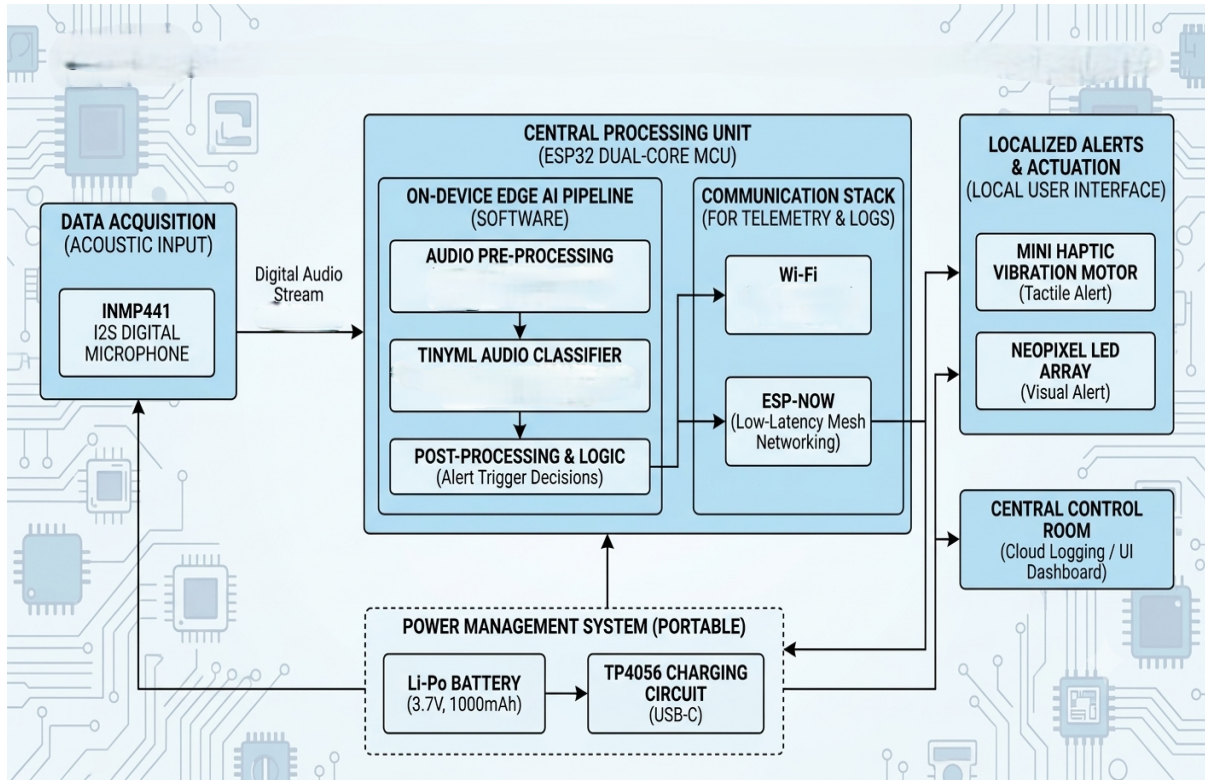
Block Diagram

Figure 2.1: Block Diagram

Chapter 3

Feasibility And Need

Need

- **Reliance on Auditory Cues:** Safety in daily environments depends heavily on hearing critical warnings like alarms, sirens, and sudden acoustic hazards.
- **Ineffectiveness of Traditional Alerts:** Standard auditory warnings fail to alert Deaf and Hard of Hearing (DHH) individuals, creating severe safety risks.
- **Visual Fatigue & Out-of-Sight Risks:** Relying solely on constant visual scanning causes mental fatigue and leaves individuals vulnerable to hazards outside their line of sight, leading to delayed emergency responses or accidents.

Feasibility

- **Local Processing with Edge AI & TinyML:** Recent advances allow complex sound processing to run locally on low-power, compact hardware without needing a continuous cloud connection.
- **Accessible Hardware Integration:** Assistive wearable nodes can easily integrate quantized neural networks onto embedded systems using digital MEMS microphones and dual-core microcontrollers.
- **Ultra-Low Latency & Cost Efficiency:** Micro-acoustic anomalies can be evaluated on-device to trigger immediate haptic alerts at a reduced operational cost, eliminating the need for expensive medical implants.

Chapter 4

Methodology And Planing Work

4.0.1 Work Breakdown Structure (WBS)

1. **Phase 1: Problem Definition & Requirement Analysis (Aug-Sep):** Literature review on assistive acoustic technologies for DHH, finalizing system specifications, component selection (ESP32-S3, INMP441, DRV2605L), and defining project scope.
2. **Phase 2: Custom PCB Layout & EDA Simulation (Oct–Nov):** Schematic entry, custom two-layer PCB artwork design in KiCad/EasyEDA, circuit simulation, and exporting Gerber files for vendor manufacturing.
3. **Phase 3: Dataset Collection & Model Training (Nov–Dec):** Capturing target environmental sound classes (Fire Alarms, Vehicle Horns, Normal Background, Dog Barks), applying noise data augmentation (-15 dB to +10 dB SNR), training CNN model in TensorFlow, and performing INT8 Post-Training Quantization.
4. **Phase 4: Hardware Assembly & Embedded Firmware (Jan–Feb):** PCB soldering, setting up FreeRTOS dual-core tasks, implementing I2S DMA audio ring buffers, porting ESP-DSP MFCC feature extraction, and deploying TFLite Micro runtime.
5. **Phase 5: Enclosure Design, Integration & Testing (Feb–Mar):** Designing a 3D-printed PETG vest/shoulder clip enclosure, assembling the complete wearable node, verifying execution latency (<50 ms), and measuring classification accuracy under ambient urban and domestic noise.
6. **Phase 6: Final Documentation, Verification & Presentation (Mar–Apr):** Completing the thesis report draft, verifying experimental charts, preparing presentation slides, and demonstrating the live prototype for evaluation.

Chapter 5

Facility Required

Software Required:

1. Arduino IDE / ESP-IDF Framework
2. KiCad / EasyEDA Tools
3. LTspice / Proteus Simulation Software
4. TensorFlow Lite Micro / Edge Impulse

Hardware Required:

5. ESP32-S3 Microcontroller & INMP441 I2S MEMS Mic
6. DRV2605L Haptic Driver & LRA Actuator
7. PCB Etching Solution & Copper Clad Board
8. Soldering Gun and Desolder Material
9. Bread Board & Testing Instruments (Oscilloscope, Multimeter)
10. 3D Printer (PETG Enclosure Fabrication)

Book Required:

11. Reference books for Embedded Systems and TinyML
12. IEEE Access Research Papers
13. Data Sheets (ESP32-S3, INMP441, DRV2605L)
14. Electronic Design Books

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